

Zirui Ren

Third-year undergraduate at Department of Physics, Tsinghua University, Beijing, China
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RESEARCH INTERESTS

Mechanistic interpretability, machine learning theory, training dynamics, physics of AI, natural language processing.

EDUCATION

Tsinghua University, Department of Physics, Beijing, China

Sep. 2023 — present

Major in Physics

Minor in Computer Science and Technology

- GPA: 3.93/4.0
- Language Proficiency: TOEFL 113/120 (Reading: 30, Listening: 29, Speaking: 25, Writing: 29)

PUBLICATIONS

- **Are Your Reasoning Models Reasoning or Guessing? A Mechanistic Analysis of Hierarchical Reasoning Models**

Zirui Ren, Ziming Liu

arXiv:2601.10679. *Under review at NeurIPS 2026*

- **LLM×MapReduce-V2: Entropy-Driven Convolutional Test-Time Scaling for Generating Long-Form Articles from Extremely Long Resources**

Haoyu Wang, Yujia Fu, Zhu Zhang, Shuo Wang, **Zirui Ren**, Xiaorong Wang, Zhili Li, Chaoqun He, Bo An, Zhiyuan Liu, Maosong Sun

arXiv:2504.05732

PROJECTS

Spectral bias analysis via frequency domain dynamics

Advisor: Ziming Liu, College of AI, Tsinghua University

Jan. 2026 — Present

- Investigating the *spectral bias* phenomenon in neural networks through Fourier-domain analysis, focusing on systematic characterization of frequency-dependent convergence dynamics in visual models and mini-RL systems.
- Developing a frequency-based framework to decouple training dynamics across spectral components using controlled toy models, aiming to establish semi-quantitative laws governing high-frequency convergence.

Mechanistic analysis of hierarchical reasoning models

Advisor: Ziming Liu, Massachusetts Institute of Technology

Jul. 2025 — Jan. 2026

- Hierarchical reasoning models (HRMs) achieve extraordinary performance on various reasoning tasks, significantly outperforming CoT-based reasoners. I conducted a systematic mechanistic study on the reasoning patterns of HRMs. The paper (first author) is submitted to NeurIPS 2026.
- I identified and corrected violations of the fixed-point assumption in HRMs. By analyzing latent-state trajectories, I discovered a hidden “guessing” behavior underlying the reasoning architecture. I developed three augmentation techniques that boosted HRM performance on *Sudoku-Extreme* from 55% to 97%, based on these insights.
- I independently implemented the end-to-end research pipeline (PyTorch, distributed training, statistical analysis) for this work and maintained the open-source codebase.

Test-time scaling framework for long-to-long generation

Internship at ModelBest.

Advisor: Zhiyuan Liu, Tsinghua University

Feb. 2025 — Aug. 2025

- We developed an LLM test-time scaling framework inspired by CNN architecture, enabling automated generation of extremely long passages with extremely long accompanying source texts. Our method outperforms representative baselines in long-form coherence, structure, content quality, and precision of reference utilization. Our paper is available on arXiv.
- I contributed to the academic survey generation software SurveyGO based on our proposed algorithm. I developed code that handles incremental refinement of generated text with newly added references.

Integrating hybrid linear attention mechanism into MiniCPM models

Internship at ModelBest.

Advisor: Zhiyuan Liu, Tsinghua University

Jun. 2025 — Aug. 2025

- We conducted an ablation study of linear attention modules (1-D convolution, forgetting gate, RoPE, and other detailed settings) in a 0.6B hybrid model, with evaluation on ARC-Challenge/Easy, HellaSwag, PIQA, Lambada, and MMLU. The model was distilled from Qwen3-0.6B via a 3-stage process devised in the RADLADS paper. Subsequently, we aligned this method with MiniCPM (the flagship model of ModelBest).
- In this project, I gained hands-on experience with the pretraining, deployment (e.g., Qwen3 models), and evaluation of foundation language models. I received comprehensive training in Python/PyTorch engineering.

SELECTED COURSES

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| • Quantum Mechanics: A+ | • Data Structures: A |
| • Electrodynamics: A | • Complex Analysis: A |
| • Basic Topology: A | • Numerical Analysis: A- |
| • Linear Algebra: A+ | • Optimization Theory: A |
| • Mathematical Physics Equations: A | • Linear Regression Analysis: A |

AWARDS

Tsinghua Comprehensive Excellence Scholarship

Qin Xiao Scholarship (Top 5%)

Oct. 2025

Tsinghua Comprehensive Excellence Scholarship

Xi Tai Scholarship (Top 5%)

Oct. 2024

Tsinghua Academic Excellence Scholarship (Twice)

Oct. 2024 & Oct. 2025

TECHNICAL SKILLS

Programming

Python (PyTorch, NumPy), C/C++ (performance-critical implementations), CUDA.

Machine Learning

LLM pretraining/distillation (0.6B-7B scale), test-time scaling methods.

Theoretical Foundations

Deep reinforcement learning, optimization theory, numerical analysis, data structures.